SOLUTION

SECTION–I

1. Ans. (C)

$P^2 = I - P$

$P^3 = P - P^2 = P - I + P = 2P - I$

$P^4 = 2P^2 - P = 2(I - P) - P = 2I - 3P$

$P^5 = 2P - 3P^2 = 2P - 3(I - P)$

$= 5P - 3I$

$P^6 = 5P^2 - 3P = 5(I - P) - 3P = 5I - 8P$

$\therefore n = 6$

2. Ans.(D)

$$\Delta = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$$

$C_1 + C_2 \to C_1$

$$\Delta = \begin{vmatrix} 2 & \cos^2 x & \sin 2x \\ 2 & 1+\cos^2 x & \sin 2x \\ 1 & \cos^2 x & 1+\sin 2x \end{vmatrix}$$

$R_1 - R_2 \to R_1$

$$\Delta = \begin{vmatrix} 0 & -1 & 0 \\ 2 & 1+\cos^2 x & \sin 2x \\ 1 & \cos^2 x & 1+\sin 2x \end{vmatrix}$$

$\Delta = 2 + \sin 2x$

Maximum value = 3 minimum value = 1

$\alpha = 3, \beta = 1, \alpha + \beta^{99} = 4, \alpha^3 - \beta^{17} = 26$

$\alpha^{2n} - \beta^{2n}$ is always even but triangle with side

$\alpha = 3, \beta = 1$ & $5\alpha - 13\beta = 2$ is not possible.

3. Ans. (A)

$C_2 \to C_2 - C_3$

$$\begin{vmatrix} a_1a_2 & d & a_0 \\ a_2a_3 & d & a_1 \\ a_3a_4 & d & a_2 \end{vmatrix}$$

$R_3 \to R_3 - R_2$ & then $R_2 \to R_2 - R_1$

$$\begin{vmatrix} a_1a_2 & d & a_0 \\ a_2(2d) & 0 & d \\ a_3(2d) & 0 & d \end{vmatrix} = -2d^3(a_2 - a_3) = 2d^4$$

4. Ans. (C)

trace (adj(|A||B| AB))

$= \text{trace}(|A|^{n-1}|B|^{n-1}(\text{AdjB})(\text{AdjA}))$

trace(Adj(|AB|BA))

$= \text{trace}(|A|^{n-1}|B|^{n-1}(\text{AdjA}).(\text{AdjB}))$

5. Ans. (A)

$\because BC = I$

$\therefore B^2C^2 = I$

$A(B^2C^2)A^T = AA^T$

which is a symmetric matrix.

6. Ans. (C)

$2A + B = B^T - A^T$ ...(i)

take transpose both side

$2A^T + B^T = B - A$ ...(ii)

equation (i) + equation (ii)

$\Rightarrow 3A = -3A^T \quad \Rightarrow A^T = -A$

$\Rightarrow 2A + B = B^T + A \quad \Rightarrow A + B = B^T$

$\Rightarrow |A + B| = |B^T| = |B| = 1$

7. Ans. (A)

$|\text{adj}A| = 3^{16} \Rightarrow |A|^{n-1} = 3^{16}$

$\Rightarrow 3^{2(n-1)} = 3^{16} \Rightarrow n = 9$

$\Rightarrow$ number of elements $= n^2 = 81$

8. Ans. (D)

$R_3 \to R_3 + R_2$

$$\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \cos\left(\theta+\frac{\pi}{6}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & -\sin\left(2\theta+\frac{\pi}{3}\right) \\ -\sin\theta & -\cos\theta & -\sin 2\theta \end{vmatrix}$$

so determinant is identically zero as two rows are identical $\Rightarrow$ It will have infinite solutions

9. Ans. (B,C,D)

$$D = \begin{vmatrix} 1 & 1 & a \\ 2 & 3 & 0 \\ 3 & 4 & a^2 \end{vmatrix} = a(a-1)$$

for $a = 0 \Rightarrow D = 0, D_x = 0, D_y = 0, \quad D_z = 2 - b$

so for $a = 0, b = 2 \Rightarrow$ Infinite Solutions.

for $a = 0, b = 1 \Rightarrow$ No solutions.

for $a = 1, D = 0 \Rightarrow D_x = 0, D_y = 0, D_z = 0$

$\Rightarrow$ Infinite solutions.

If $a \neq 0$ but $a = 1 \Rightarrow$ does't have unique solutions.

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10. Ans. (B,C)

Characteristic equation of A is

$x^3 - 2x^2 - 4x + 1 = 0$

$\Rightarrow \lambda = -4$

$\because A^3 - 2A^2 - 4A + I = 0$

$\therefore A^2 + A^{-1} = 2A + 4I$

$\Rightarrow |A^2 + A^{-1}| = |2A + 4I| = 56$

Paragraph for Question 11 to 12

11. Ans. (C)

$\because |A^T|A|B| = |A|B|B^T|$

$\therefore |A|\,|A|\,|B| = |B|\,|A|\,|B|$

$\Rightarrow |A| = 2$

$\Rightarrow |AB^{-1}\ \text{adj}\ (A^TB)^{-1}|$

$$= \frac{|A|}{|B|}\frac{1}{|A^T|^2|B|^2} = \frac{1}{16}$$

$$= \frac{1}{|A||B|^3} = \frac{1}{2\times 2^3} = \frac{1}{16}$$

12. Ans. (C)

$B = B^T$

option (A) $\text{adj}(B) = \text{adj}(B^T)$

option (B) $\text{adj}(B^{-1})^T = (B^T)^{-1} = B^{-1}$

option (C) $A^T|A|B = A|B|B^T$

$\Rightarrow A^TB = AB^T$

$\Rightarrow A^TB = AB$ or $A = A^T$

but $A^{2015}B^{2016}$ is symmetric if $AB = BA$

option (D) $(AB^{-1})^T = (B^{-1})^T A^T = B^{-1}A^T = B^{-1}A = AB^{-1}$ ($\because AB = BA$)

Paragraph for Questions 13 to 15

13. Ans. (A)

Constant term (absolute term) will be same as g(0)

since $g(y) = Ay^3 + By^2 + Cy + D$

$g(0) = (p + q + r)(p^2 + q^2 + r^2 - pq - qr - rp) > 0$

14. Ans. (D)

$g'(y) = 3Ay^2 + 2By + C$

$\Rightarrow C = g'(0)$

$g'(0) = C = 3qr - 3p^2$

15. Ans. (A)

$y_1 = 1,\ y_2 = w,\ y_2 = w^2$

$g(0) = p^3 + q^3 + r^3 - 3pqr$

$= (p + q + r)(p + qw + rw^2)(p + qw^2 + rw)$

$\Rightarrow g(0) = f(y_1) \cdot f(y_2) \cdot f(y_3)$

SECTION–III

1. Ans. 7.00

$$\begin{vmatrix} 1 & \sin\alpha & \sin^2\alpha \\ 1 & \cos\alpha & \cos^2\alpha \\ 1 & \sin 2\alpha & \sin^2 2\alpha \end{vmatrix} = 0$$

$\Rightarrow (\sin\alpha - \cos\alpha)(\cos\alpha - \sin 2\alpha)(\sin 2\alpha - \sin\alpha) = 0$

$\Rightarrow \tan\alpha = 1,\ \cos\alpha = 0,\ \sin\alpha = \frac{1}{2},\ \sin\alpha = 0,$

$\cos\alpha = \frac{1}{2}$

$\Rightarrow \alpha = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{5\pi}{6}$ & π

2. Ans. 6.00

Put x = 1

$a_4 + a_3 + a_2 + a_1 + a_0$

$$= \begin{vmatrix} 2 & -1 & 1 \\ 5 & 1 & 3 \\ -4 & -1 & 2 \end{vmatrix} = 10 + 22 - 1 = 31$$

3. Ans. 25.00

$$A_2 = \begin{vmatrix} a & b+3c & c+4a \\ b & c+3a & a+4b \\ c & a+3b & b+4c \end{vmatrix} + 2\begin{vmatrix} b & b+3c & c+4a \\ c & c+3a & a+4b \\ a & a+3b & b+4c \end{vmatrix}$$

Assume $A_2 = B_1 + 2B_2$

In B_1 : $C_3 \to C_3 - 4C_1$ & In B_2 : $C_2 \to C_2 - C_1$

$$A_2 = \begin{vmatrix} a & b+3c & c \\ b & c+3a & a \\ c & a+3b & b \end{vmatrix} + 6\begin{vmatrix} b & c & c+4a \\ c & a & a+4b \\ a & b & b+4c \end{vmatrix}$$

Assume $A_2 = B_3 + 6B_4$

In B_3 : $C_2 \to C_2 - 3C_3$ & In B_4 : $C_3 \to C_3 - C_2$

$$A_2 = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} + 6\begin{vmatrix} b & c & 4a \\ c & a & 4b \\ a & b & 4c \end{vmatrix}$$

$$= \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} + 24\begin{vmatrix} b & c & a \\ c & a & b \\ a & b & c \end{vmatrix} = 25A_1$$

4. Ans. 162.00

$\because$ Trace of matrix is 1 $\Rightarrow$ diagonal elements are 1,1,–1 or 1,0,0

$\therefore$ Number of possible symmetric matrices

$= (3 \times 3^3) \times 2 = 162$

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